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AI-Driven Early Warning and Intervention Support System for At-Risk Higher Education Students

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1. LITERATURE REVIEW

1.1. Early Prediction At-Risk Students

Early academic risk prediction uses Machine Learning to identify students who may face poor academic performance before their final results. Previous research has used academic results, attendance and Learning Management System (LMS) activities for this purpose. Riestra-González et al. (2021) analyzed around 8.5 million LMS records from 15,994 students across 669 courses. Their Multilayer Perceptron achieved 80.1% accuracy at 10% course completion and 90.1% at 50%, showing that students can be identified as at-risk during early stages. Gaftandzhieva et al. (2022) analyzed Moodle activities and attendance of 105 students and found that four weeks of data was insufficient for reliable prediction, while eight weeks produced better results. However, the small sample limits its generalizability. Santos and Henriques (2023) further investigated course-agnostic risk prediction using LMS logs. Ewais et al. (2026) used 1,419 student records and twelve academic features, where Random Forest achieved an R^2 of 0.93, although R^2 cannot be directly compared with classification accuracy. However, existing studies commonly rely on specific data sources such as LMS activity or limited student samples. Accuracy alone can also hide poor identification of the smaller at-risk group, making recall and F1-score important for evaluation. This creates a gap for risk prediction using multiple academic and behavioral indicators together. Therefore, this component will combine attendance, assessment performance, previous GPA, LMS activity and missed assignments to identify a student's current risk as Low, Medium, or High.

1.2. Explainable Student Academic Risk Factors

Existing work in Educational Data Mining focuses on combining Explainable Artificial Intelligence (XAI) with predictive models to create useful tools for teachers. The importance of having interpretable models was indicated by Alamri and Alharbi (2021). They showed the need for explainable solutions. Following that, Nagy and Molontay (2023) used pre-enrollment data together with XAI techniques to predict which STEM students were likely to become dropouts. Following that, Jang et al. (2022) worked alongside teachers to demonstrate that simple Logistic Regression models with visuals were able to successfully predict academic achievement during the first weeks of studies. Likewise, active learner behaviour can be analysed through deep learning and XAI to understand which factors affect the students' risk, as shown by Kala et al. (2024). Islam et al. (2025) showed that virtual learning settings can be combined with machine learning and XAI to study learner interaction with LMS and their test results. Overall, all of these approaches help to turn complex algorithms' outputs into easy-tounderstand reports. Current research focuses more on forecasting and understanding student academic risk but not necessarily on providing immediate intervention to address that risk.

1.3. Temporal Analysis of Student Behaviors and Academic Performance

Recent learning analytics research shows that analyzing student behavior over time can help identify academic difficulties earlier. Chen and Cui (2020) showed that changes in LMS activity can help predict course performance. Similarly, Akçapınar et al. (2019) analyzed student usages at different stages of a course to identify students at risk. Omarbekova et al. (2026) used a Bi-LSTM with temporal attention to record and analyze student activity over 32 weeks, while Van, Tieu, and Tran (2026) combined student characteristics with time-series data, including online activity and assessment performance. Zhang and Duan (2026) also examined changes in student activity and found that declining engagement was associated with increased academic risk. These studies demonstrate the value of monitoring student behavior over time. However, existing approaches do not fully address the need to classify changes across multiple academic indicators into clear trajectories, such as improving, stable, or deteriorating, within an early-warning system. For example, a student may maintain LMS engagement but have attendance or assessment performance declining. Therefore, this component will analyze long term student data, including attendance, assessment performance, LMS engagement, and missed submissions, to identify these different patterns of change. This will help educators recognize students whose academic situation is deteriorating, even when their current performance alone may not indicate an immediate risk.

1.4. Evidence-Based Student Intervention and Outcome Monitoring

Beyond the average effects of academic support reviewed by Sønderlund et al. (2019), treatment-effect studies estimate which students benefit: Beemer et al. (2017) used a stacked ensemble for supplemental instruction, Smith et al. (2020) validated a machine-learning method against propensity score matching and added SHAP, and Xiao et al. (2024) estimated heterogeneous effects from 48 trials. Action-recommendation studies link prediction to advice: Albreiki et al. (2022) used explainable ML with rules, Afzaal et al. (2021) gave LMS-based feedback, and Alalawi et al. (2025) linked predictions to study plans and reported a higher at-risk pass rate. However, each covers one intervention or setting, or derives actions from a student's current profile, and none reuses tracked intervention outcomes as evidence for later recommendations. This component addresses that gap.

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2. RELATED WORK SUMMARY

Research	Author	Year	Dataset	Model	Metric
Early Prediction of At-Risk Students					
Massive LMS Log Data Analysis for the Early Prediction of Course-Agnostic Student Performance	Riestra-González et al.	2021	15,994 students, 669 courses, ~8.5 million LMS log records from the University of Oviedo	Decision Tree, Naive Bayes, Logistic Regression, SVM, MLP; K-Means for behavioral analysis	MLP accuracy: 80.1% when 10% of the course had elapsed; 90.1% at 50%.
Exploring Online Activities to Predict the Final Grade of Student	Gaftandzhieva et al.	2022	105 OOP students; 7,057 Moodle activity records and 738 online lecture attendance records	Statistical techniques and multiple ML classifiers	Random Forest accuracy: 71% on the imbalanced full dataset and 78% on the balanced full dataset. Four weeks of data could not reliably identify low performers; eight weeks provided moderate prediction.
Accurate, Timely, and Portable: Course-Agnostic Early Prediction of Student Performance from LMS Logs	Santos and Henriques	2023	LMS interaction/log data; also evaluated using data from other institutions to test portability	Course-agnostic ML models using static-aggregate and time-dependent LMS representations	At-risk prediction: AUC 0.756 using all course data. High-performing student prediction: 78.2% accuracy using Extremely Randomized Tree

Research	Author	Year	Dataset	Model	Metric
Using Machine Learning for Early Identification of At-Risk Students	Ewais et al.	2026	Real-world higher-education dataset; 1,419 students and twelve academic features	Linear Regression, Ridge, Decision Tree, Random Forest, Gradient Boosting, SVR	Random Forest Regressor: $R^2 = 0.93$. This is a regression result, so do not compare it directly with classification accuracy.
Explainable Student Academic Risk Factors					
The integration of explainable AI in Educational Data Mining for student academic performance prediction and support system	Islam et al.	2025	Benchmark UCI Student Performance Dataset	Decision Tree, Random Forest, Gradient Boosting, and XGBoost SHAP (Shapley Additive explanations), LIME, Shapash, and ELI5	XGBoost / Random Forest): Achieved 92%96% classification accuracy
Interpretable Dropout Prediction: Towards XAI-Based Personalized Intervention	Nagy and Molontay	2023	8,508 student records from Budapest University of Technology and Economics, Hungary	Cat-Boost Classifier, Natural Gradient Boosting (NG-Boost), Logistic Regression, Explainable AI (XAI)	CatBoost Classifier: ROCAUC of 0.8859 and a Macro F1Score of 0.9143
Explainable Student Performance Prediction Models: A Systematic Review	Alamri and Alharbi	2021	Systematic literature corpus of primary educational data mining (EDM) studies (2015-2020)	Ante-hoc (inherently interpretable/white-box models)	Interpretability vs. Accuracy Trade-off:. (Decision Trees, Rule Mining at 70–80%), Accuracy Range Across Literature ranging between 70% and 95%.

Research	Author	Year	Dataset	Model	Metric
Practical early prediction of students' performance using machine learning and explainable AI	Jang et al.	2022	University course datasets (U1 and U2 datasets)	Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), XGBoostSHAP (Shapley Additive explanations)	Logistic Regression (Optimal Early Classifier): Achieved 71.88%–75.0% Accuracy and strong ROCAUC performance
Early Prediction of Student Performance in Face-to-Face Education Environments: A Hybrid Deep Learning Approach With XAI Techniques	Kala et al.	2024	Face-to-face university educational dataset.	Hybrid Deep Learning framework (integrating Convolutional Neural Networks with Recurrent/Gated Neural Networks)	Hybrid Deep Learning Model Accuracy: 89.3% Precision: 96.5% Recall: 90.5% F1-Score: 93.4%
Temporal Analysis of Student Behavior and Academic Performance					
Utilizing Student Time Series Behaviors in Learning Management Systems for Early Prediction of Course Performance	Chen and Cui	2020	LMS student activity data	LSTM, compared with eight conventional ML classifiers	Education course: Best validation AUC was 75.2% using the first 56 days. Using selected features, Naive Bayes achieved 66.2% validation AUC. Biology course: Best validation AUC was 83.0% using the first 70 days. Using selected features, Naive Bayes achieved 67.3% validation AUC.

Research	Author	Year	Dataset	Model	Metric
Using learning analytics to develop early-warning system for at-risk students	Akçapınar et al.	2019	Online learning interaction data from seventy-six university students	kNN and other classification algorithms	kNN: correctly identified 89% of unsuccessful students using end-of-term data. Using week 3 data, it identified 74% of the students who failed
A temporal attention-based hybrid deep learning model for student performance and academic risk prediction	Omarbekova et al..	2026	OULAD	Bi-LSTM + MLP + Temporal Attention	ROC-AUC 0.9497, weighed F1 0.9008 and accuracy 0.9024; recall for the at-risk group was 0.79.
A multimodal learning analytics model based on deep learning for predicting student performance using tabular and time-series data fusion	Van et al.	2026	OULAD + self-collected dataset	MLP + LSTM + Aggregation, compared with LSTM, RF etc.	on OULAD, MSE 0.2404, MAE 0.3413 and $R^2 = 0.7836$using three data sources. On the self-collected dataset, MSE 0.7410, MAE 0.6636 and $R^2 = 0.2893$using four sources.
Artificial intelligence-driven early warning for online learner dropout: temporal explainability and theory-grounded intervention	Zhang and Duan	2026	OULAD — 32,593 registrations	Six ML algorithms; LightGBM performed best/joint-best + Temporal SHAP	0.911 (91.1%) for both LightGBM and XGBoost at T4. At T3 (day 90), LightGBM achieved AUC 0.890, F1 0.731, precision 0.781, recall 0.687 and AUC-PR 0.814. In prospective evaluation, AUC was 0.731 at T3, showing lower performance when predicting future withdrawals among students still enrolled.

Research	Author	Year	Dataset	Model	Metric
Evidence-Based Student Intervention and Outcome Monitoring					
Ensemble Learning for Estimating Individualized Treatment Effects in Student Success Studies	Beemer et al.	2017	Introductory statistics course records, San Diego State University (1,032 students)	Stacked generalization, ridge regression, ensemble ML	Ensemble RMSE 45.3 (final exam, out of three hundred) and 0.887 (course grade); accuracy 80.52%, AUC 0.82 (grade C or better); average ITE +9.3 exam points
Global and Individual Treatment Effects Using Machine Learning Methods	Smith et al.	2020	Engineering mechanics course data with additional tutorials	ML, propensity score matching, counterfactual prediction, SHAP	Mean absolute error of 6% to 8% when predicting final grades
Uncovering Individualized Treatment Effects for Educational Trials	Xiao et al.	2024	Forty-eight educational randomized controlled trials, over 200,000 students	Multilevel linear model (best), GLM, random forest; causal inference	Mean squared error 0.42 for the best model, versus 0.49 (GLM) and 0.50 (random forest)
Explainable AI for Data-Driven Feedback and Intelligent Action Recommendations to Support Students' Self-Regulation	Afzaal et al.	2021	LMS data from a programming course, Stockholm University (157 students analyzed)	Random forest, MLP, BayesNet, KNN, SVM, LR; explainable ML; dashboard	Random forest accuracy 0.75 (assignment prediction) and 0.91 (quiz prediction)

Research	Author	Year	Dataset	Model	Metric
Framework for Automatically Suggesting Remedial Actions to Help Students at Risk	Albreiki et al	2022	Student performance data (at risk = total grade below 70%)	Explainable ML, rule-based models	reached 1.00 for a binary classifier using four checkpoints. This unusually high result should be discussed cautiously and checked against the evaluation design and possibility of leakage.
Evaluating the Student Performance Prediction and Action Framework Through a Learning Analytics Intervention Study	Alalawi et al.	2025	Historical student assessment data; large undergraduate course	Predictive modelling, learning analytics	Binary at-risk accuracy 0.93 (KNN, after project 2). Pass rate of at-risk students 45.5% with the intervention versus 10.9% in the control cohort ($p < 0.0001$)

3. OVERALL GAP

Even though current studies have analyzed the early risk prediction of academic performance, the ability of explainable AI, predicting student behavior changes over time, and interventions that can be offered, these areas are studied in isolation from each other. Because of this situation, it is necessary to design an integrated model that combines all mentioned capabilities and helps at-risk students in higher education institutions more effectively. In order to fill in the identified gap, the first step is to predict which students would be at risk and what factors explain their risk level. The second aim is to identify changes in their academic behavior over time. Finally, we should define interventions that can be offered to these students and monitor the effectiveness of the proposed methods. By combining these four elements, the integrated system will allow educators to detect academic problems earlier and choose more adequate interventions.

4. SUMMARY

The aim of our project is to develop a smart early warning and student assistance system that identifies and helps students in higher education who could be in danger of subpar academic performance. The system performs analysis on academic, behavioral, and engagement data to show current academic risk, specify its causes, follow changes in student behavior over time, and suggest necessary interventions based on existing data. Each element aimed to provide educators with information based on data so that they can identify at-risk students at an early stage.

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